

Quick overview:

This protocol is to evaluate the effectiveness of the clamping mechanism used in the Link V1 Thigh actuator modules. These modules feature custom electrical connections for transmitting power and EtherCAT communication throughout the rest of the device. This will provide insight on the robustness of the design and effectiveness of the electrical connections. This experiment should inform if the custom connections are suitable for the Link V1 device and other iterations in the future.

Goals:

Understand the characteristics and performance continuity of the electrical connections during a number of stress tests that would otherwise be influenced by assisted tasks by the device.

Conditions:

1. Compression/Tension test
2. Wave test
3. Bashing test
4. Vibration test (?)

Materials:

- Dual actuator module packages linked to a central CF thigh bar
- Nuc
- Power supply
- Beckhoff modules
- Vibration instrument (?)

Setup:

- Check data logger and logger camera to make sure they're running properly
- Attach necessary electronics to both ends of the thigh link
- Attach cabling between the Mk1 exo Nuc and ECAT Line

Experiment:**Compression/Tension**

- Place the thigh link vertically onto the work bench, ensuring there are no electrical wires being pinched
- Apply constant pressure from the top end of the link downward onto the table
- Perform the same for tension

Wave

- Suspend the thigh link in the air, ensuring there is enough cable slack on either side of the link
- Rapidly swing, shake, and jostle the thigh

Bashing

- Hold the thigh link vertically with respect to the work bench, while holding electrical wires aside
- Hammer the link down onto the workbench repetitively (10 cycles max)

- Hold the thigh link horizontally with respect to the work bench, while holding electrical wires aside
- Hammer the distal end of the link onto the edge of the workbench repetitively (10 cycles max)

Vibration

- Procure instrument capable of oscillating motion or vibration. Set to maximum setting if possible
- Hold instrument directly against either end of the thigh link, preferably onto the clamping feature
- Adjust instrument position gradually across the outer profile of the actuator module

Tear down:

- Unplug all cables from both ends of the thigh link
- Inspect connections and clamped surfaces on both sides

Notes: